

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

KALAHSTHI RAMYA
192311149RA

CO3-AT3-Debugging and Optimization

[← Back](#)

QUESTIONS

Q1

ATM Withdrawal – Exception
Handling Debugging
10 marks

Q2

Hospital Registration –
NullPointerException
Debugging
10 marks

Q3

E-Commerce Product Search –
Array Exception Debugging
10 marks

Q4

Bank Account – Race Condition
Debugging
10 marks

Q5

Railway Reservation –
Synchronization Debugging
10 marks

5/5 answered

Q3

E-Commerce Product Search – Array
Exception Debugging10
marks

An online shopping application allows customers to select a product using a product number from 1 to 5.

The following program contains an array indexing error.

Task:

1. Identify the error.
2. Correct the product-number and array-index logic.
3. Handle invalid product numbers.
4. Handle non-numeric input.
5. Submit the COMPLETE corrected Java program.
6. The program must display "Search completed" after processing.

Faulty Program:

```
import java.util.Scanner;

class ProductSearch {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String[] products = {
            "Laptop",
            "Mobile",
```

```

"Tablet",
"Headphone",
"Keyboard"
};

System.out.print("Enter product number (1-5): ");
int productNumber = sc.nextInt();

System.out.println("Selected product: "
+ products[productNumber]);

System.out.println("Search completed");
}
}

```

Correct the program so that product numbers 1, 2, 3, 4 and 5 correctly access the corresponding products.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 import java.util.Scanner;
2 import java.util.InputMismatchException;
3
4 class Main{
5     public static void main(String[] args){
6         Scanner sc = new Scanner(System.in);
7
8         String[] products = {"Laptop","Mobil
9
10        try{
11            System.out.print("Enter product
12            if(!sc.hasNextInt()){
13                throw new InputMismatchExcep
14            }
15
16            int productNumber = sc.nextInt()
17
18            if(productNumber < 1 || productN
19                throw new ArrayIndexOutOfBou
20        }

```

```
21         int targetIndex = productNumber
22         System.out.println("Selected Pro
23     }catch(InputMismatchException e){
24         System.out.println("Error: "+e.g
25     }catch(ArrayIndexOutOfBoundsException
26         System.out.println("Error: "+e.g
27     }finally{
28         System.out.println("Search compl
29         sc.close();
30     }
31 }
32 }
```

Test Input

laptop

Output

Enter product number (1 to 5): Error: Invalid
input! Please enter a number.
Search completed

🕒 Pending manual review

✅ Submitted